



James
Wijaya

Bachelor in Engineering Science

MECHATRONICS TRACK

PROFILE

Bachelor in Engineering Science focusing on modeling, control, and mechatronic systems. Experience in embedded programming and mechanical design through academic and personal projects, with an interest in robotics and autonomous systems.

CONTACT

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[GitHub : James Wijaya](#)

LANGUAGES

English: C1 (TOEFL 108, 13/12/2025)
French: B2+ (DELTA B2, 06/12/2022)
Indonesian: Native
German: Beginner A2

TECHNICAL SKILLS

Programming & Control Software:
Python, C/C++, MATLAB / Simulink,
Linux, PLC Siemens
Embedded Systems:
Arduino IDE, Quartus Prime (FPGA)
Design Software:
Autodesk Inventor, Altium Designer
FreeCAD, OnShape
Office Tools:
Microsoft Office Suite, LaTeX

OTHER SKILLS

Workplace First Aid Training (SST)

EDUCATION

Bachelor in Engineering Sciences – Mechatronics, 2025 – 2026

University of Strasbourg, Strasbourg, France.

High School Diploma in Mathematic and Natural Sciences, 2020–2023

Ignatius Global School, Palembang, Indonesia.

INTERNSHIP

Laboratory Intern, June – August 2026

University of Strasbourg–Icube, Strasbourg, France.

> Fixed-wing drone modelling and simulation using MATLAB/Simulink

Electrical Installer Intern, May – June 2025

SANTERNE Alsace – Civil Hospital Site, Strasbourg, France.

> Interpretation of electrical plans and technical documentation

> Installation of power supply and lighting networks

PROJECTS

6-DOF Bipedal Robot (ongoing)

University of Strasbourg, Strasbourg, France.

> Designed and assembled a 6-DOF bipedal robotic platform using CAD tools and additive manufacturing (PLA-based structure)

> Performed structural and power feasibility analysis with respect to actuator constraints and load transmission

> Evaluated center of mass and support polygon for stability assessment

> Initiated kinematic modeling for future control design and motion analysis

Greenhouse Control Model (embedded system implementation)

University of Strasbourg, Strasbourg, France.

> Designed and implemented an embedded control system for temperature and humidity regulation using an ATmega32U4 microcontroller

> Developed real-time sensor acquisition and signal processing pipeline

> Implemented bang-bang control with hysteresis for stable temperature regulation around a configurable setpoint (18–28 °C range)

> Evaluated preliminary P/PI control strategies and conducted simplified thermal system identification

ORGANISATION ACTIVITIES

Treasurer, Sept. 2025 – Present

Public Relations Committee Member, June 2024 – August 2025

Indonesian Student Association in Alsace (PPI Alsace).

> Financial management and budgeting

> Event logistics coordination and member communication